Optimization Frameworks for Hybrid Biomass Power-to-Methanol System: MILP and Reinforcement Learning Approaches

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Statement of Originality

I hereby certify that the work embodied in this thesis is the result of original research, is free of plagiarised materials, and has not been submitted for a higher degree to any other University or Institution.

Aug. 2018	Signature
....	....
Date	Muhammad Rafi Afif Indrajaya

Supervisor Declaration Statement

I have reviewed the content and presentation style of this thesis and declare it is free of plagiarism and of sufficient grammatical clarity to be examined. To the best of my knowledge, the research and writing are those of the candidate except as acknowledged in the Author Attribution Statement. I confirm that the investigations were conducted in accord with the ethics policies and integrity standards of Nanyang Technological University and that the research data are presented honestly and without prejudice.

Mar. 2019	Signature
....	....
Date	Prof. Einstein XXX

Acknowledgements

I wish to express my greatest gratitude to my advisor.

"If I had one hour to save the world, I would spend 55 minutes defining the problem and only five minutes finding the solution."

—Einstein, Albert

To my dear family

Abstract

My abstracts

Contents

List of Figures

List of Tables

Symbols and Acronyms

Symbols

$\mathcal{R}^n$	the n-dimensional Euclidean space
$\mathcal{H}$	the Euclidean space
$\|\cdot\|$	the 2-norm of a vector or matrix in Euclidean space
$\|\cdot\|_G$	the induced norm of a vector in G-space
$\|\cdot\|_E$	the induced norm of a vector or matrix in probabilistic space
$\odot$	the Hadamard (component-wise) product
$\otimes$	the Kronecker product
$\langle\cdot,\cdot\rangle$	the inner product of two vectors
$\circ$	the composition of functions
∇f	the gradient vector
$\mathcal{C}^k$	the function with continuous partial derivatives up to k orders
$x_{i,k}$	the i-th component of a vector x at time k
$\bar{x}$	the vector with the average of all components of x as each element
$\mathbf{1}$	all-ones column vector with proper dimension
$\mathcal{C}$	the average space, i.e., $span\{\mathbf{1}\}$
$\mathcal{C}^\perp$	the disagreement space, i.e., $span^\perp\{\mathbf{1}\}$
$\Pi_\parallel$	the projection matrix to the average space $\mathcal{C}$
$\Pi_\perp$	the projection matrix to the disagreement space $\mathcal{C}^\perp$
$O(\cdot)$	order of magnitude or ergodic convergence rate (running average)
$o(\cdot)$	non-ergodic convergence rate
$\mathcal{N}_i$	the index set of the neighbors of agent i

Acronyms

DOP	Distributed Optimization Problem
EDOP	Equivalent Distributed Optimization Problem
SDOP	Stochastic Distributed Optimization Problem
OEP	Optimal Exchange Problem
OCP	Optimal Consensus Problem
DOCP	Dynamic Optimal Consensus Problem
AugDGM	Augmented Distributed Gradient Methods
AsynDGM	Asynchronous Distributed Gradient Methods
D-ESC	Distributed Extremum Seeking Control
D-SPA	Distributed Simultaneous Perturbation Approach
D-FBBS	Distributed Forward-Backward Bregman Splitting
ADMM	Alternating Direction Method of Multipliers
DSM	Distributed (Sub)gradient Method
GAS	Globally Asymptotically Stable
UGAS	Uniformly Globally Asymptotically Stable
SPAS	Semi-globally Practically Asymptotically Stable
USPAS	Uniformly Semi-globally Practically Asymptotically Stable
HoS	Heterogeneity of Stepsize
FPR	Fixed Point Residual
OBE	Objective Error
i.i.d.	independent and identically distributed
a.s.	almost sure convergence of a random sequence

Chapter 1

Introduction

1.1 Background

- The importance of industrial-scale energy transition motivates solutions for hard-to-abate industries through Power-to-X.
- This complex industrial process, especially with the unpredictability of renewable energy sources, is financially intensive and requires optimization strategies to minimize investment and operational dispatch costs.
- The current robust conventional optimization strategy, MILP, is the industrial standard for optimizing these processes. However, the rise of AI applications across industries has motivated researchers to explore machine learning, specifically reinforcement learning, to optimize dispatch strategy as an agentic plant manager.
- It is also evident that MILP simplifies complex systems into linear representatives, which can reduce result accuracy. One example is hydrogen tank operation, where the relation between hydrogen level and efficiency is non-linear due to compression energy requirements at higher tank levels. Therefore, this thesis also observes the effect of integrating this detail into the proposed system and its impact on the techno-economic results produced by the RL agent.

- This level of detail and complexity is not possible to integrate in a MILP problem, which can be a significant limitation for a complex polygeneration system with crucial non-linear behavior across components and reactions.

- Most available studies currently focus on comparing these two frameworks in simple energy systems with fewer than four decision variables. This study pushes that boundary by observing results with roughly twice as many decision variables.

- In addition, other studies combine control and design optimization into one optimization problem, whereas this study adopts a two-step outerloop and innerloop approach to obtain combined optimized sizing and dispatch results.

This thesis compares the techno-economic results produced by an RL agent against a conventional MILP framework for a hybrid biomass power-to-methanol plant in South Africa across different scenarios and case studies. The goal is to understand the role of RL in complex Power-to-X industrial processes and benchmark its current performance against an industrial-standard framework.

1.2 Introduction to Company

- This master's thesis research is conducted at FEV Europe GmbH in the energy+resources department.

- FEV Europe GmbH is a leading expert in automotive industries, known for benchmarking and consulting services, and is now supporting innovation in the energy sector.

- This thesis supports identification of an efficient optimization tool for innovative energy solutions in complex polygeneration plants for future FEV Europe GmbH projects.

1.3 Objective

- Run and analyze the two optimization frameworks under different scenarios.

- Evaluate whether a non-linear physical model with an RL agent converges to significantly different component sizing/layout than MILP, and whether this changes techno-economic performance metrics such as CAPEX, OPEX, and LCOM.
- Identify reasons behind discrepancies, or lack thereof, depending on simulation outcomes.
- Conclude whether one framework is better overall or whether suitability is scenario-dependent.

Expected impact:

- Better framework selection for hybrid methanol synthesis plants.
- Easier optimal-framework identification for other industrial systems.
- Increased accuracy or reduced effort for similar optimization processes.
- Better understanding of where and how RL agents should be applied in polygeneration systems.

1.4 Thesis Configuration

- Chapter 2 discusses state-of-the-art research on RL for optimizing complex industrial processes or power systems, and introduces related comparative studies from the literature.
- Chapter 3 presents the steps used to compare these optimization frameworks. The comparison is conducted in multiple stages and explained in detail.
- Chapter 4 presents multistage comparison results for different scenarios and case studies, and introduces a generalization scoring approach to compare framework performance across aspects.
- Chapter 5 summarizes general findings while acknowledging limitations, assumptions, and considerations adopted in this thesis.

Chapter 2

Literature Review

2.1 Power-to-X Integration for Methanol Synthesis System

Different definitions of linearity and non-linearity for methanol synthesis systems are discussed in the literature.

2.2 MILP for Complex Polygeneration System

This section is kept as a placeholder and will be expanded with detailed MILP-specific discussion.

2.3 State-of-the-art RL in Optimizing Energy System and Industrial Processes

The adoption of RL to optimize energy systems and industrial processes has attracted significant attention due to flexibility and adaptability [1]. By directly interacting with the system, an RL agent derives strategies through a Markov Decision Process (MDP). This process involves dynamic evolution of states, actions, transition probabilities, and rewards at each time step.

- Diagram of MDP process.

The main objective of an agent is to maximize the discounted sum of reward from time t to infinity.

$$R_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \tag{2.1}$$

The agent navigates actions using a policy π, which determines what to do at each state. Each policy has an associated state-value function $V^\pi(s)$.

$$V^\pi(s) = \mathbb{E}_\pi[R_t \mid s_t = s] \tag{2.2}$$

The optimal value function is defined over all policies.

$$V^*(s) = \max_\pi V^\pi(s) \tag{2.3}$$

Another related variable is the action-value function $Q^\pi(s, a)$.

$$Q^\pi(s, a) = \mathbb{E}_\pi[R_t \mid s_t = s, a_t = a] \tag{2.4}$$

Its optimal form is defined as follows.

$$Q^*(s, a) = \max_\pi Q^\pi(s, a) \tag{2.5}$$

In short, an optimal policy π^* is one whose state-value function equals the optimal state value.

$$\pi^* = \arg\max_\pi V^\pi(s) \tag{2.6}$$

- Diagram of these variables as an MDP process (to be recreated from literature).

Accurate environment dynamics modeling through MDPs is emphasized by many researchers to obtain robust frameworks and good results [2]. This supports RL use for complex and dynamic environments such as polygeneration plants.

- Examples of RL agents used in energy-system optimization.

Different RL algorithm families are commonly used for complex energy-system optimization: value-based, policy-based, and actor-critic. They differ mainly by learning mechanism. Value-based RL learns a value estimate, commonly $Q(s, a)$, then chooses actions with highest estimated value. Typical examples include Q-learning and DQN [3]. A related study argues value-function methods can face convergence issues in continuous multivariate settings, where they are less suitable than in discrete cases [4].

Policy-based RL learns the policy directly, parameterizing the action rule and searching for an optimal policy, including methods such as policy gradients and PPO [5]. Because many energy problems are continuous, high-dimensional, and constrained, policy-based approaches are often preferred [6].

Actor-critic methods combine both views: the actor learns policy and the critic learns value estimates to stabilize training. Earlier literature adopted Q-learning more frequently, while recent work increasingly applies actor-critic and other deep RL variants to complex control tasks [7]. Common algorithms include PPO, SAC, A3C, TD3, and DDPG.

2.4 RL vs MILP for Different Applications

Recent literature provides several RL-vs-MILP benchmarks for energy systems. Vetter et al. [8] compared DQN-based RL against MILP for dispatch optimization in a Grid-PV-TES energy community system with fixed sizing. RL controlled TES SOC, while MILP optimized detailed power flows. Both reduced annual cost versus baseline, with MILP showing larger cost reduction. RL, however, performed better on some additional metrics such as CO_2 reduction, self-consumption ratio, and self-sufficiency ratio because it tended to consume more PV generation.

Beyond dispatch-only benchmarking, Cauz et al. [9] compared RL and MILP under both design and control decisions in Battery-PV systems. The RL algorithm used was Direct Environment and Policy Search (DEPS). Design decisions were battery and PV sizing; control decisions were battery and grid dispatch. In stage one (one-week case), both approaches converged under fixed sizing. Under joint design-control optimization, RL initially appeared cheaper, but this was traced to modeling limitations (missing end-SOC consistency). After correcting assumptions, MILP became cheaper, highlighting the importance of matched assumptions for fair comparison.

In stage two (one-year case), RL remained more expensive than MILP under both fixed-sizing and joint design-control scenarios, while producing substantially different design choices. The study attributes this to multiple causes, including hyperparameter sensitivity and algorithm selection.

Chapter 3

Methodology

3.1 Hybrid Biomass Power-to-Methanol System Model

- Energy system definition.
- Technical parameters definition.
- Economical parameters definition.

3.2 Two-steps Outerloop & Innerloop Strategy

The methodology adopts a two-step strategy in which layout/sizing and operational dispatch decisions are decomposed into outerloop and innerloop problems to improve tractability and interpretability.

3.3 Level 2 Optimization Architecture (MILP)

- Component-level non-linear modeling logic.
- Operation logic.
- Modeling tool: `oemof.solph`.

- Solver: CBC.
- Objective function definition.

3.4 Level 3 Optimization Architecture (RL)

- Component-level linear modeling logic.
- Chosen RL algorithm.
- Operation logic (MDP).
- Integration of Bayesian Optimization (BO) and RL.
- Objective function definition (reward function).

3.5 Data

This section is reserved for dataset sources, preprocessing, assumptions, and scenario-specific inputs.

3.6 Comparison Stages & Scenario Definition

This comparison is conducted in two steps:

- **Outerloop comparison**
  - Innerloop operational decision is fixed as an `oemof` problem.
  - For outerloop (layout selection fed into innerloop), different tools are compared: `oemof` investment block (IB), Bayesian Optimization (BO), and Design of Experiment (DOE).
  - Objective: identify which approach gives the best layout under both linear and non-linear economies of scale.
  - Components to optimize: electricity supply and Power-to-X blocks.

- **Innerloop comparison**
  - Compare RL dispatch in a physics-based plant model against `oemof` dispatch in a linear model.
  - Stage 1 (fixed outerloop, linear model with RL): evaluate how close RL can optimize a linear model compared to Level 2 MILP.
  - Stage 2 (fixed outerloop, non-linear model with RL): evaluate behavior when optimized linear layouts are injected into a physics-based model, including feasibility and performance changes.
  - Stage 3 (joint outerloop + innerloop): compare BO+RL non-linear pathway vs BO+`oemof` linear pathway.

Case studies for each step (especially innerloop stage comparisons):

- **Weather variability**
  - Stable solar and wind input.
  - Extreme weather volatility.
- **Market conditions**
  - Cheap batteries.
  - Cheap hydrogen system (e.g., -50%, -20%, +20%, +50%).
- **System design and flexibility**
  - Heavy fixed methanol reactor load.
  - Heavy fixed electrolyzer load.

For each step, stage, and case study, predetermined KPIs are examined:

- **Techno-economic level**
  - LCOM.
  - CAPEX and OPEX.
  - Carbon intensity.

  - Grid dependency ratio.
  - Self-consumption ratio.
  - Autonomy days.

- **Framework level**
  - Simulation time.
  - BO distance.

- **Scenario descriptors**
  - Year.
  - CAPEX and OPEX per component.
  - Conversion factor per component.

- **Fixed components**
  - MeOH reactor.
  - Distillery.
  - MeOH storage.
  - Anaerobic digester.
  - ATR.
  - O_2 storage (full ATR-load sizing for 3 days).
  - Steam generator.
  - Auxiliary units: RWGS, fuel boiler, AD gas separator, compressor, and three heat exchangers.

- **Variable components (IB vs BO vs DOE)**
  - CH_4 generator or gas turbine.
  - Wind turbine.
  - PV.
  - BESS.
  - Electrolyzer.
  - Hydrogen storage.
  - Biogas storage.

Chapter 4

Results and Discussions

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Chapter 5

Conclusions

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5.1 Section 1

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Appendix A

Proofs for Part I or Chapter 3

A.1 Proof of Lemma

$$\psi^{av}(\theta) = \frac{1}{T}\int_0^T [\psi(\theta + \mu(\tau)) + C] \otimes \frac{\mu(\tau)}{a} d\tau$$

A.2 Proof of another Lemma

$$\begin{aligned} \gamma_1(\|x\|) \leq W(t,x) \leq \gamma_2(\|x\|) \\ \frac{\partial W}{\partial t} + \frac{\partial W}{\partial x}\phi(t,x,0) \leq -\gamma_3(\|x\|) \end{aligned} \tag{A.1}$$

List of Author's Awards, Patents, and Publications[1]

Awards

- **Best Paper Awards**, "A Great System," *Nature*.

Patents

- **A Great System**, "A Great System," *Nature*.

Journal Articles

- **My name** and My colleague, "A Great System," *Nature*.

Conference Proceedings

- **My name**, My colleague 1, My colleague 3 and My colleague 3, "Greater System," in *Conference of Vision, 2018*.

[1] The superscript * indicates joint first authors

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